



Title

rdbwselect — Bandwidth Selection Procedures for Local Polynomial Regression Discontinuity Estimators.

Syntax

```
rdbwselect depvar indepvar [if] [in] [, c(#) fuzzy(fuzzyvar [sharpbw]) deriv(#)
p(#) q(#) covs(covars) covs_drop(covsdropoption) kernel(kernelfn)
weights(weightsvar) bwselect(bwmethod) all scaleregul(#)
masspoints(masspointsoption) bwcheck(bwcheck) bwrestrict(bwropt)
stdvars(stdopt) precision(precopt) vce(vce_type [vceopt1 vceopt2]) ]
```

Description

rdbwselect implements bandwidth selectors for local polynomial Regression Discontinuity (RD) point estimators and inference procedures developed in [Calonico, Cattaneo and Titiunik \(2014a\)](#), [Calonico, Cattaneo and Farrell \(2018\)](#), [Calonico, Cattaneo, Farrell and Titiunik \(2019\)](#), and [Calonico, Cattaneo and Farrell \(2020\)](#).

Companion commands are: [rdrobust](#) for point estimation and inference procedures, and [rdplot](#) for data-driven RD plots (see [Calonico, Cattaneo and Titiunik \(2015a\)](#) for details).

A detailed introduction to this command is given in [Calonico, Cattaneo and Titiunik \(2014b\)](#), and [Calonico, Cattaneo, Farrell and Titiunik \(2017\)](#). A companion R package is also described in [Calonico, Cattaneo and Titiunik \(2015b\)](#).

Related Stata and R packages useful for inference in RD designs are described in the following website:

<https://rdpackages.github.io/>

Requires Stata 16 or later.

Options

Estimand

c(#) specifies the RD cutoff for *indepvar*. Default is **c**(0).

fuzzy(fuzzyvar [*sharpbw*]) specifies the treatment status variable used to implement fuzzy RD estimation (or Fuzzy Kink RD if **deriv**(1) is also specified). Default is Sharp RD design and hence this option is not used. If the option *sharpbw* is set, the fuzzy RD estimation is performed using a bandwidth selection procedure for the sharp RD model. This option is automatically selected if there is perfect compliance at either side of the threshold.

deriv(#) specifies the order of the derivative of the regression functions to be estimated. Default is **deriv**(0) (for Sharp RD, or for Fuzzy RD if **fuzzy**(.) is also specified). Setting **deriv**(1) results in estimation of a Kink RD design (up to scale), or Fuzzy Kink RD if **fuzzy**(.) is also specified.

Local Polynomial Regression

p(#) specifies the order of the local polynomial used to construct the point estimator. Default is **p**(1) (local linear regression).

q(#) specifies the order of the local polynomial used to construct the bias correction. Default is **q**(2) (local quadratic regression).

covs(covars) specifies additional covariates to be used for estimation and inference.

covs_drop(*covsdropoption*) assess collinearity in additional covariates used for estimation and inference. Options **pinv** (default choice) and **invsym** drops collinear additional covariates, differing only in the type of inverse function used. Option **off** omits the check for collinear additional covariates.

kernel(*kernelfn*) specifies the kernel function used to construct the local-polynomial estimator(s). Options are: **triangular**, **epanechnikov**, and **uniform**. Default is **kernel(triangular)**.

weights(*weightsvar*) is the variable used for optional weighting of the estimation procedure. The unit-specific weights multiply the kernel function.

Bandwidth Selection

bwselect(*bwmeth*) specifies the bandwidth selection procedure to be used. Options are:

mserd one common MSE-optimal bandwidth selector for the RD treatment effect estimator.

msetwo two different MSE-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator.

msesum one common MSE-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof).

msecomb1 for min(**mserd**,**msesum**).

msecomb2 for median(**msetwo**,**mserd**,**msesum**), for each side of the cutoff separately.

cerd one common CER-optimal bandwidth selector for the RD treatment effect estimator.

certwo two different CER-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator.

cersum one common CER-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof).

cercomb1 for min(**cerd**,**cersum**).

cercomb2 for median(**certwo**,**cerd**,**cersum**), for each side of the cutoff separately.

Note: MSE = Mean Square Error; CER = Coverage Error Rate.

Default is **bwselect(mserd)**. For details on implementation see [Calonico, Cattaneo and Titiunik \(2014a\)](#), [Calonico, Cattaneo and Farrell \(2018\)](#), [Calonico, Cattaneo, Farrell and Titiunik \(2019\)](#), and [Calonico, Cattaneo and Farrell \(2020\)](#), and the companion software articles.

all if specified, **rd**wselect**** reports all available bandwidth selection procedures.

scaleregul(#) specifies scaling factor for the regularization term added to the denominator of the bandwidth selectors. Setting **scaleregul(0)** removes the regularization term from the bandwidth selectors. Default is **scaleregul(1)**.

masspoints(*masspointsoption*) checks and controls for repeated observations in the running variable. Options are:

off ignores the presence of mass points.

check looks for and reports the number of unique observations at each side of the cutoff.

adjust controls that the preliminary bandwidths used in the calculations contain a minimal number of unique observations. By default it uses 10 observations, but it can be manually adjusted with the option **bwcheck**.

Default option is **masspoints(adjust)**.

bwcheck(*bwcheck*) if a positive integer is provided, the preliminary bandwidth used in the calculations is enlarged so that at least *bwcheck* unique observations are used.

bwrestrict(*bwropt*) if set **on**, computed bandwidths are restricted to lie within the range of *runvar*. Default is **on**.

stdvars(*stdopt*) if set **on**, *depvar* and *runvar* are standardized before computing the bandwidths. Standardization avoids numerical instability in bandwidth selection when *runvar* has very large or very small magnitude. Default is **on**.

precision(*precopt*) controls the storage precision of internal temporary variables (cluster-id, mass-point counts). Options are **double** (default) and **single**; **single** maps to Stata's **float** storage type. Default is **precision(double)**.

Variance-Covariance Estimation

vce(*vcetype* [*vceopt1* *vceopt2*]) specifies the procedure used to compute the variance-covariance matrix estimator. Options are:

- vce**(**nn** [*nnmatch*]) for heteroskedasticity-robust nearest neighbor variance estimator with *nnmatch* indicating the minimum number of neighbors to be used.
- vce**(**hc0**) for heteroskedasticity-robust plug-in residuals variance estimator without weights.
- vce**(**hc1**) for heteroskedasticity-robust plug-in residuals variance estimator with *hc1* weights.
- vce**(**hc2**) for heteroskedasticity-robust plug-in residuals variance estimator with *hc2* weights.
- vce**(**hc3**) for heteroskedasticity-robust plug-in residuals variance estimator with *hc3* weights.
- vce**(**cluster** *clustervar*) for cluster-robust plug-in residuals variance estimation with degrees-of-freedom weights; equivalent to **vce**(**cr1** *clustervar*).
- vce**(**cr1** *clustervar*) for cluster-robust plug-in residuals variance estimator with degrees-of-freedom correction.
- vce**(**cr2** *clustervar*) for the Bell-McCaffrey (2002) bias-reduced cluster-robust variance estimator (CRV2).
- vce**(**cr3** *clustervar*) for the Pustejovsky-Tipton (2018) cluster-robust variance estimator (CRV3), approximately unbiased with few clusters.

The CR2/CR3 leverage correction applies to both the conventional and the robust bias-corrected standard errors, including when the point-estimation bandwidth *h* differs from the bias-correction bandwidth *b*; in that case the cluster leverage is computed from the bias (*b*) regression.

Default is **vce**(**nn** 3).

Example: Cattaneo, Frandsen and Titiunik (2015) Incumbency Data

```

Setup
. use rdrobust_senate.dta

MSE bandwidth selection procedure
. rdbwselect vote margin

All bandwidth bandwidth selection procedures
. rdbwselect vote margin, all

```

Stored results

rdbwselect stores the following in **e()**:

```

Scalars
  e(N)           number of observations
  e(N_l)         number of observations to the left of the cutoff
  e(N_r)         number of observations to the right of the cutoff
  e(c)           cutoff value
  e(p)           order of the polynomial used for estimation of the
                  regression function
  e(q)           order of the polynomial used for estimation of the bias of
                  the regression function estimator
  e(n_clust)     number of clusters (only when vce(cluster|cr1|cr2|cr3 ...)
                  is specified)

```

```

e(h_mserd)      MSE-optimal bandwidth selector for the RD treatment
                 effect estimator.
e(h_msetwo_1)   MSE-optimal bandwidth selectors below the cutoff for the
                 RD treatment effect estimator.
e(h_msetwo_r)   MSE-optimal bandwidth selectors above the cutoff for the
                 RD treatment effect estimator.
e(h_msesum)     MSE-optimal bandwidth selector for the sum of regression
                 estimates.
e(h_msecomb1)   for min(mserd,msesum) .
e(h_msecomb2_1) for median(msetwo,mserd,msesum), below the cutoff.
e(h_msecomb2_r) for median(msetwo,mserd,msesum), above the cutoff.

e(h_cerrd)      CER-optimal bandwidth selector for the RD treatment
                 effect estimator.
e(h_certwo_1)   CER-optimal bandwidth selectors below the cutoff for the
                 RD treatment effect estimator.
e(h_certwo_r)   CER-optimal bandwidth selectors above the cutoff for the
                 RD treatment effect estimator.
e(h_cersum)     CER-optimal bandwidth selector for the sum of regression
                 estimates.
e(h_cercomb1)   for min(cerrd,cersum) .
e(h_cercomb2_1) for median(certwo_1,cerrd,cersum), below the cutoff.
e(h_cercomb2_r) for median(certwo_r,cerrd,cersum), above the cutoff.

e(b_mserd)      MSE-optimal bandwidth selector for the bias of the RD
                 treatment effect estimator.
e(b_msetwo_1)   MSE-optimal bandwidth selectors below the cutoff for the
                 bias of the RD treatment effect estimator.
e(b_msetwo_r)   MSE-optimal bandwidth selectors above the cutoff for the
                 bias of the RD treatment effect estimator.
e(b_msesum)     MSE-optimal bandwidth selector for the sum of regression
                 estimates for the bias of the RD treatment effect
                 estimator.
e(b_msecomb1)   for min(mserd,msesum) .
e(b_msecomb2_1) for median(msetwo,mserd,msesum), below the cutoff.
e(b_msecomb2_r) for median(msetwo,mserd,msesum), above the cutoff.

e(b_cerrd)      CER-optimal bandwidth selector for the bias of the RD
                 treatment effect estimator.
e(b_certwo_1)   CER-optimal bandwidth selectors below the cutoff for the
                 bias of the RD treatment effect estimator.
e(b_certwo_r)   CER-optimal bandwidth selectors above the cutoff for the
                 bias of the RD treatment effect estimator.
e(b_cersum)     CER-optimal bandwidth selector for the sum of regression
                 estimates for the bias of the RD treatment effect
                 estimator.
e(b_cercomb1)   for min(cerrd,cersum) .
e(b_cercomb2_1) for median(certwo_1,cerrd,cersum), below the cutoff.
e(b_cercomb2_r) for median(certwo_r,cerrd,cersum), above the cutoff.

```

Macros

```

e(cmd)          rdbwselect
e(cmdline)      command as typed
e(title)        title (RD bandwidth selection (sharp/fuzzy), optionally ,
                 covariate-adjusted)
e(depvar)       name of dependent (outcome) variable
e(runningvar)   name of running variable
e(clustvar)     name of cluster variable
e(covs)         names of additional covariates
e(vce_select)   vcetype specified in vce() (raw option name, e.g. "nn",
                 "hc3", "cr3")
e(vce_type)     display label for variance-covariance estimator (e.g.
                 "NN", "HC3", "CR3")
e(bwselect)     bandwidth selection choice
e(kernel)       kernel choice
e(precision)    storage precision selected via precision(): double
                 (default) or single

```

Matrices

e(mat_h) 1x2 matrix of bandwidths for the RD treatment effect estimator (left, right)

e(mat_b) 1x2 matrix of bandwidths for the bias of the RD treatment effect estimator (left, right)

References

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